

Microeconomic Theory: TA Session

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Reminder

In-class quiz during next Friday's TA session

Friday Sep 25th, 9:00-9:50am, Gilman 132

Closed-book quiz; scientific or “four-function”
calculators allowed

Homework 2, Q1

There are two goods, peanut butter (x) and jelly (y). Johns starts with 1 unit of x and 7 units of y , which we write as $(e_x^J, e_y^J) = (1, 7)$, and has utility function $u_J(x, y) = x^2y$. Hopkins starts with $(e_x^H, e_y^H) = (9, 3)$, and has utility function $u_H(x, y) = x/2 + y$.

(b) [In (a) we show that another allocation can Pareto dominate the endowment.] Explain why each can be made better off through exchange.

Answer: Johns has convex preferences, meaning he prefers more balanced bundles of peanut butter and jelly (presumably because he eats them together). Hopkins, however, regards 2 units of peanut butter as a perfect substitute for 1 unit of jelly (maybe he spreads either one on his crackers, but does not care which). Because Johns starts out with a relatively “jelly-heavy” endowment, he would like to trade some jelly to get more peanut butter. Hopkins is not opposed to the idea, provided he receives at least 1 unit of Jelly for every 2 units of peanut butter he trades away.

Homework 2, Q3

For goods x and y , Alice has utility function $u(x, y) = xy - 2y$.
Currently, the price per unit of x is 6 and the price per unit of y is 1. Her budget is 102.

1. Are Alice's preferences monotonic? Insatiable?

Answer: Her preferences are not monotonic. E.g. $(1, 0) \succ (1, 2)$.
But they're insatiable. For any (x, y) , we have $(x + 1, y) \succ (x, y)$

2. What are her optimal quantities of x and y ?

Answer: $MRS = \frac{p_x}{p_y} \Rightarrow \frac{y}{x-2} = 6 \Rightarrow y = 6x - 12$. Plugging into
BC: $6x + y = 6x + (6x - 12) = 102 \Rightarrow x = 9.5, y = 45$

3. What percentage of her budget is allocated to x ?

Answer: $\frac{9.5 \times 6}{102} \approx 55.9\%$.

4. If Alice's budget increases to 1200, what are her optimal quantities?

Answer: Similar to (2),
 $6x + y = 6x + (6x - 12) = 1200 \Rightarrow x = 101, y = 594$

Homework 2, Q3

For goods x and y , Alice has utility function $u(x, y) = xy - 2y$. Currently, the price per unit of x is 6 and the price per unit of y is 1. Her budget is 102.

5. What percentage of her budget is allocated to x now? Explain how and why this differs from your answer in (3).

Answer: $\frac{101 \times 6}{1200} = 50.5\%$.

As Alice's budget increases, the share of her budget spent on x approaches 50%. This is because, as her budget increases and she consumes more x and y , the $(-2y)$ term in her utility function becomes less important — the behavior of her utility function approaches that of $u = xy$.

We now show that the consumer with utility function $u = xy$ spend 50% of her income on x . We have $MRS = \frac{p_x}{p_y} \Rightarrow \frac{y}{x} = \frac{p_x}{p_y}$

$\Rightarrow y = \frac{p_x}{p_y}x$. Plugging into the budget constraint:

$p_x x + p_y y = p_x x + p_y \left(\frac{p_x}{p_y} x \right) = 2p_x x = I$. So, the share of her budget spent on x is $\frac{p_x x}{I} = \frac{p_x x}{2p_x x} = 50\%$.

Income and substitution effects

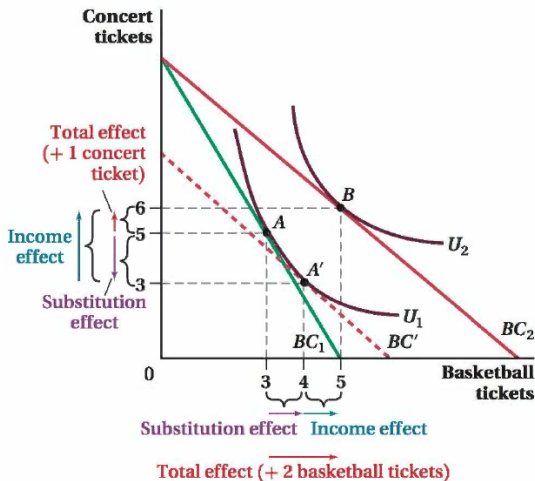


Figure: Income and substitution effects of normal goods

The “interim bundle” A' is at the old utility level but new prices

Income and substitution effects

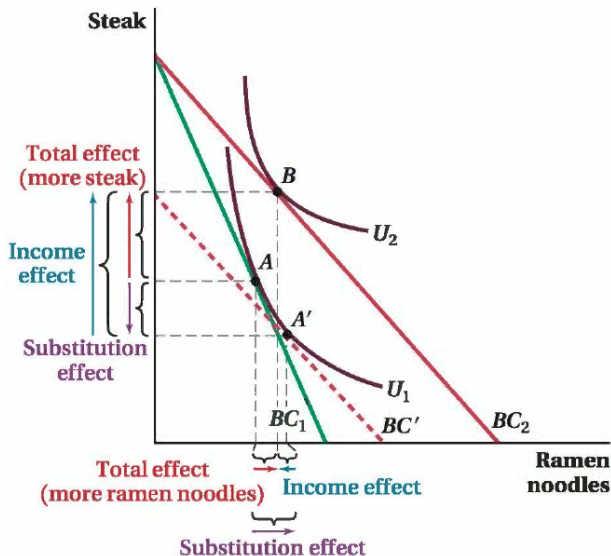


Figure: Income and substitution effects of an inferior good (x-axis)

IE and SE - exercise

Jack's utility function for goods x and y is $U = 2xy + 10x$. He has \$100 dollars. Each unit of x costs \$5, and each unit of y costs \$4.

1. Calculate his optimal consumption bundle.

Answer: $MRS = \frac{\partial U / \partial x}{\partial U / \partial y} = \frac{2y+10}{2x} = \frac{y+5}{x}$. The price ratio is $\frac{p_x}{p_y} = \frac{5}{4}$.

At the optimum, $MRS = \frac{p_x}{p_y} \Rightarrow \frac{y+5}{x} = \frac{5}{4} \Rightarrow y = \frac{5}{4}x - 5$.

Plugging this into the budget constraint: $5x + 4y = 5x + 4(\frac{5}{4}x - 5) = 10x - 20 = 100$. We have the optimal consumption bundle $x_{old} = 12$, $y_{old} = 10$.

2. Now the price of good y falls to \$1. Calculate his optimal consumption bundle.

Answer: $MRS = \frac{p_x}{p_y} \Rightarrow \frac{y+5}{x} = 5 \Rightarrow y = 5x - 5$. Plugging into the budget constraint, we have:

$5x + y = 5x + (5x - 5) = 10x - 5 = 100$. The optimal consumption bundle is $x_{new} = 10.5$, $y_{new} = 47.5$.

IE and SE - exercise

Jack's utility function for goods x and y is $U = 2xy + 10x$. He has \$100 dollars. Each unit of x costs \$5, and each unit of y costs \$4.

3. Calculate the income and substitution effects of both goods resulting from this change.

Answer: We first calculate the optimal consumption bundle under the old utility level and the new prices. The old utility level is

$$U_{old} = 2xy + 10x = 2 \times 12 \times 10 + 10 \times 12 = 360. \text{ For the}$$

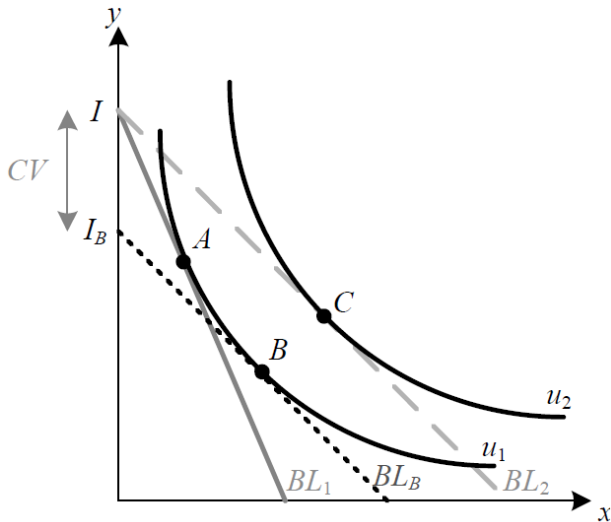
$MRS = \frac{p_x}{p_y}$ relationship under the new prices, we have

$$\frac{y+5}{x} = 5 \Rightarrow y = 5x - 5. \text{ Plugging into the utility function, we have}$$
$$2x(5x - 5) + 10x = 360 \Rightarrow x_{interim} = 6, y_{interim} = 25.$$

The substitution effect for x is $x_{interim} - x_{old} = -6$, the income effect for x is $x_{new} - x_{interim} = 4.5$.

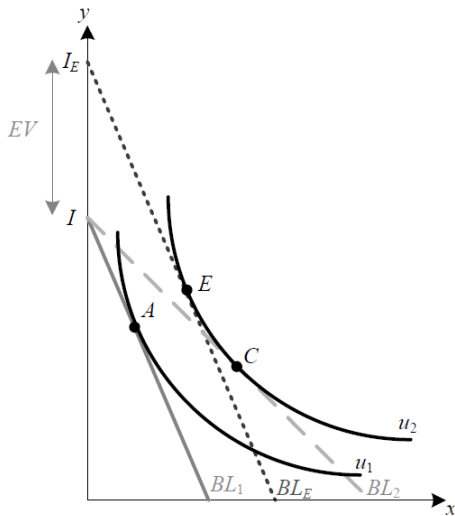
The substitution effect for y is $y_{interim} - y_{old} = 15$, the income effect for y is $y_{new} - y_{interim} = 22.5$.

Compensating variation



“old utility, new prices”

Equivalent variation



“new utility, old prices”

CV and EV - exercise

Jack's utility function for goods x and y is $U = 2xy + 10x$. He has \$100 dollars. Each unit of x costs \$5, and each unit of y costs \$4.

4. Calculate the compensating variation from a decrease in the price of y to \$1.

Answer: From question (3), we know that $x_{interim} = 6$, $y_{interim} = 25$. The budget constraint under the new prices is
 $5x + y = 5 \times 6 + 25 = 55 \Rightarrow CV = 100 - 55 = 45$

5. Calculate the equivalent variation from a decrease in the price of y to \$1.

Answer: From question (2), we know that $x_{new} = 10.5$, $y_{new} = 47.5$. The utility is $U = 2xy + 10x = 1102.5$. With the old prices, we have $MRS = \frac{p_x}{p_y} \Rightarrow y = \frac{5}{4}x - 5$ (see question (1) for derivations).

The equivalent bundle has

$$U = 2xy + 10x = 2x\left(\frac{5}{4}x - 5\right) + 10x = 1102.5 \Rightarrow x = 21, y = \frac{85}{4}.$$

The equivalent budget constraint is

$$5x + 4y = 5 \times 21 + 4 \times \frac{85}{4} = 190 \Rightarrow EV = 190 - 100 = 90.$$